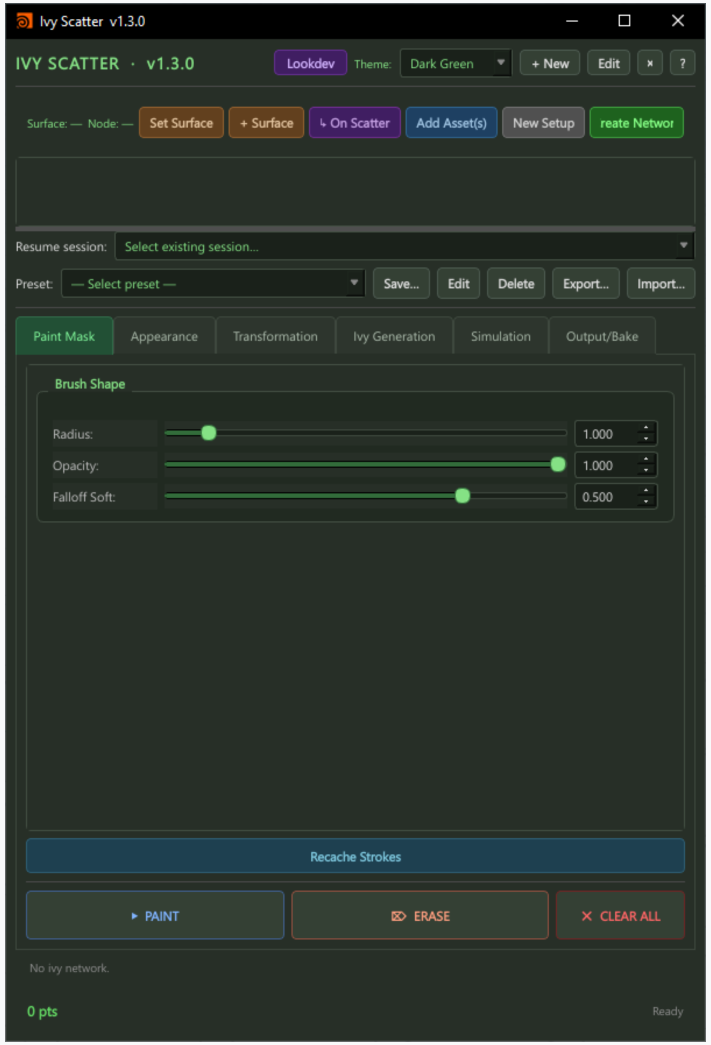
Ivy Generation

v1.3.0 · Tutorial Guide

Procedural ivy growth for Houdini · Vellum simulation · Arnold · Redshift · Karma / USD

Paint Mask Tab · Define where ivy grows by painting a density mask



Brush Shape

Radius — world-space brush size in the viewport
Opacity — stroke strength; lower values build up density gradually
Falloff Soft — 0 = hard circular edge · 1 = smooth fade at brush perimeter

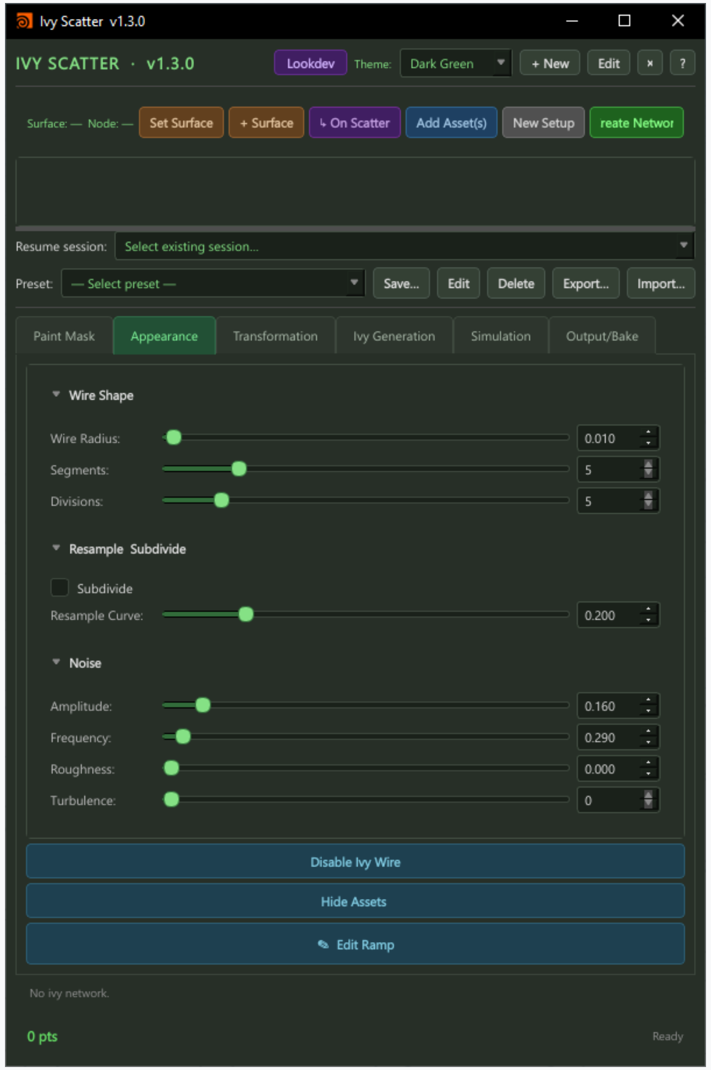
How to Use

Paint where you want ivy strands to originate on the surface
Denser painted areas produce more ivy strands
Switch to ERASE to remove painted regions locally
CLEAR ALL wipes the entire mask for this setup
Recache Strokes — force re-evaluation after changing brush parameters

Bottom Bar

PAINT — activate brush; LMB drag across surface in the viewport
ERASE — remove painted mask in brushed area
CLEAR ALL — reset the entire density mask
Status line shows current point count and cook state

Appearance Tab · Wire shape, subdivision, and surface noise for ivy strands



Wire Shape

Wire Radius — thickness of each ivy stem/wire in world units
Segments — number of cross-section polygons around each wire (higher = rounder)
Divisions — number of length-wise subdivisions along each wire strand

Resample / Subdivide

Subdivide — adds a Catmull-Clark subdivision pass to the wire geometry for smooth curves
Resample Curve — resamples the underlying growth curve at this segment length before wiring

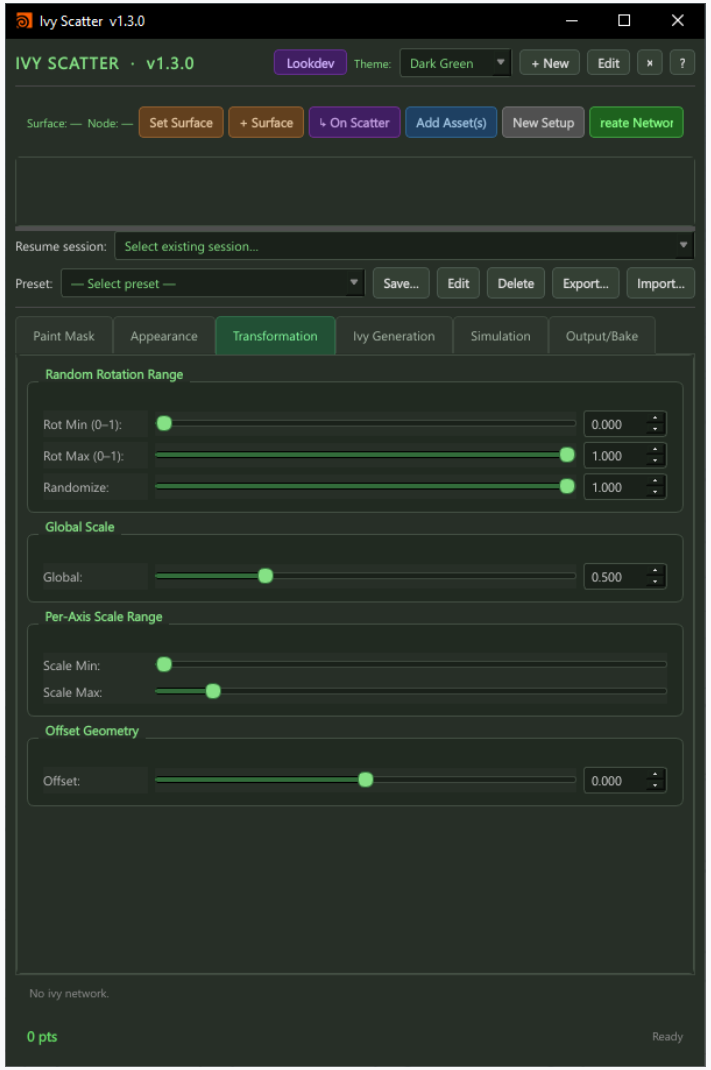
Noise

Amplitude — strength of the displacement noise applied along the wire length
Frequency — spatial frequency of the noise; higher = tighter, more frequent kinks
Roughness — adds high-frequency detail on top of the base noise
Turbulence — number of noise octaves; more octaves = more complex surface variation

Action Buttons

Disable Ivy Wire — hides wire geometry in viewport for faster interaction
Hide Assets — hides leaf/asset geometry; useful when tweaking wire shape
Edit Ramp — opens the thickness-along-length ramp for tapering wire tips

Transformation Tab · Rotation and scale variation for ivy leaf assets



Random Rotation Range

Rot Min (0–1) — minimum normalized rotation per leaf instance (0 = 0°)
Rot Max (0–1) — maximum normalized rotation per leaf instance (1 = 360°)
Randomize — blends between no variation and full Min/Max range; dial back for subtle variation

Global Scale

Global — uniform scale multiplier applied to every leaf asset instance
Use to quickly size up or down all leaves without touching per-axis ranges

Per-Axis Scale Range

Scale Min — minimum random scale per instance
Scale Max — maximum random scale per instance
Each leaf gets a unique random scale between Min and Max

Offset Geometry

Offset — moves each leaf along its surface normal by this amount
Use to lift leaves slightly off the wire surface to prevent clipping

Ivy Generation Tab · Core growth parameters — shape, gravity, and strand control

Mask Seed

- Seed — random seed for strand placement within the painted mask
- Max Strands — hard cap on the total number of growing strands
- Mask Threshold — minimum mask value required for a point to spawn a strand (0–1)

Shape

- Strand Length — total length each strand grows before stopping
- Step Size — distance advanced per growth step; smaller = smoother but slower cook
- Curl Amount — amount of curling/spiraling applied along each strand's length

Gravity Motion

- Gravity Strength — how strongly strands are pulled downward as they grow
- Droop Bias — biases gravity toward strand tips; creates realistic heavy droop
- Inertia — resistance to direction change; higher = straighter, less organic strands
- Randomness — adds stochastic variation to each growth step direction

Point Jitter

- Jitter Scale — random positional offset applied to each strand point
- Jitter Seed — seed for jitter randomisation

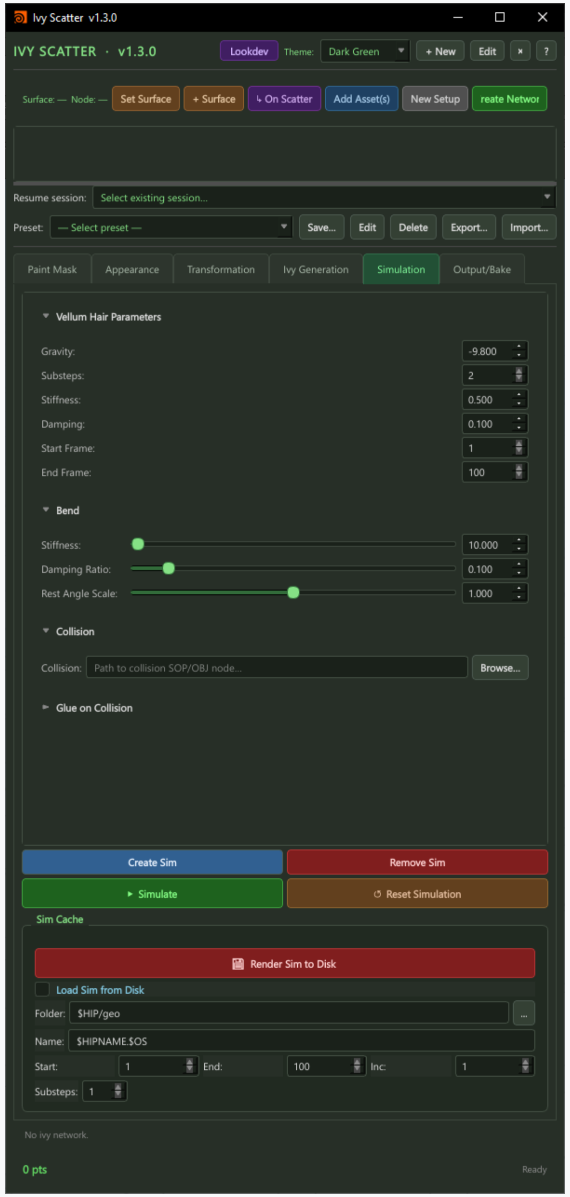
Strand Length Scale

- Min Length / Max Length — each strand gets a random length scale in this range
- Creates natural variation where not all strands grow to full length

Actions

- Presets — save, load, or update named growth configurations
- Create Ivy Network — builds the Houdini SOP network for this setup
- Remove Ivy Network — destroys the generated network nodes

Simulation Tab · Vellum hair physics for wind, droop, and collision



Vellum Hair Parameters

- Gravity — gravitational acceleration applied to strands (default −9.8)
- Substeps — simulation substeps per frame; more = more stable but slower
- Stiffness — global stiffness of the hair constraint; higher = less bending
- Damping — energy dissipation; higher = strands settle faster with less oscillation
- Start / End Frame — frame range over which the simulation runs

Bend

- Stiffness — resistance to bending; high values keep strands straight
- Damping Ratio — bend-specific damping to reduce oscillation at joints
- Rest Angle Scale — scales the natural rest angle; use to add pre-curl to simulation

Collision

- Collision — path to a SOP or OBJ node used as collision geometry
- Strands will drape over and collide with the assigned geometry
- Glue on Collision — strands stick where they first contact the collision object

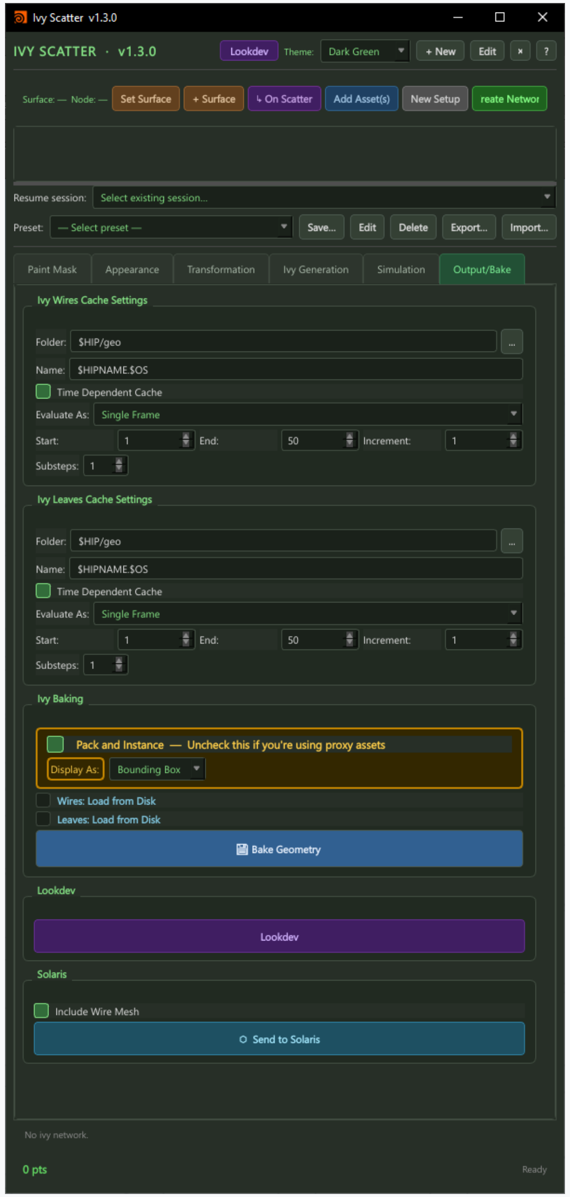
Simulation Controls

- Create Sim — builds the Vellum simulation network
- Simulate — runs the simulation for the defined frame range
- Reset Simulation — clears the cached simulation result
- Remove Sim — destroys the simulation network nodes

Sim Cache

- Render Sim to Disk — bakes the simulation result to disk cache
- Load Sim from Disk — loads previously baked simulation instead of re-simulating
- Folder / Name — cache output path and filename template

Output / Bake Tab · Cache to disk, lookdev materials, and USD / Solaris export



Ivy Wires Cache Settings

- Folder — output directory for wire geometry cache ($HIP/geo by default)
- Name — filename template ($HIPNAME.$OS)
- Time Dependent Cache — write one file per frame for animated ivy
- Evaluate As — Single Frame or Frame Range for sequence baking
- Start / End / Inc / Substeps — frame range and substep settings

Ivy Leaves Cache Settings

- Separate cache track for leaf/asset geometry
- Same settings as Wires cache — folder, name, frame range
- Wires and Leaves can be loaded independently from disk

Ivy Baking

- Pack and Instance — uncheck when using Redshift Proxy or Arnold Standin assets
- Display As — Bounding Box (fast viewport) or Full Geometry
- Wires: Load from Disk / Leaves: Load from Disk — switch between cached and live cook
- Bake Geometry — freeze the ivy geometry to disk at the paths above

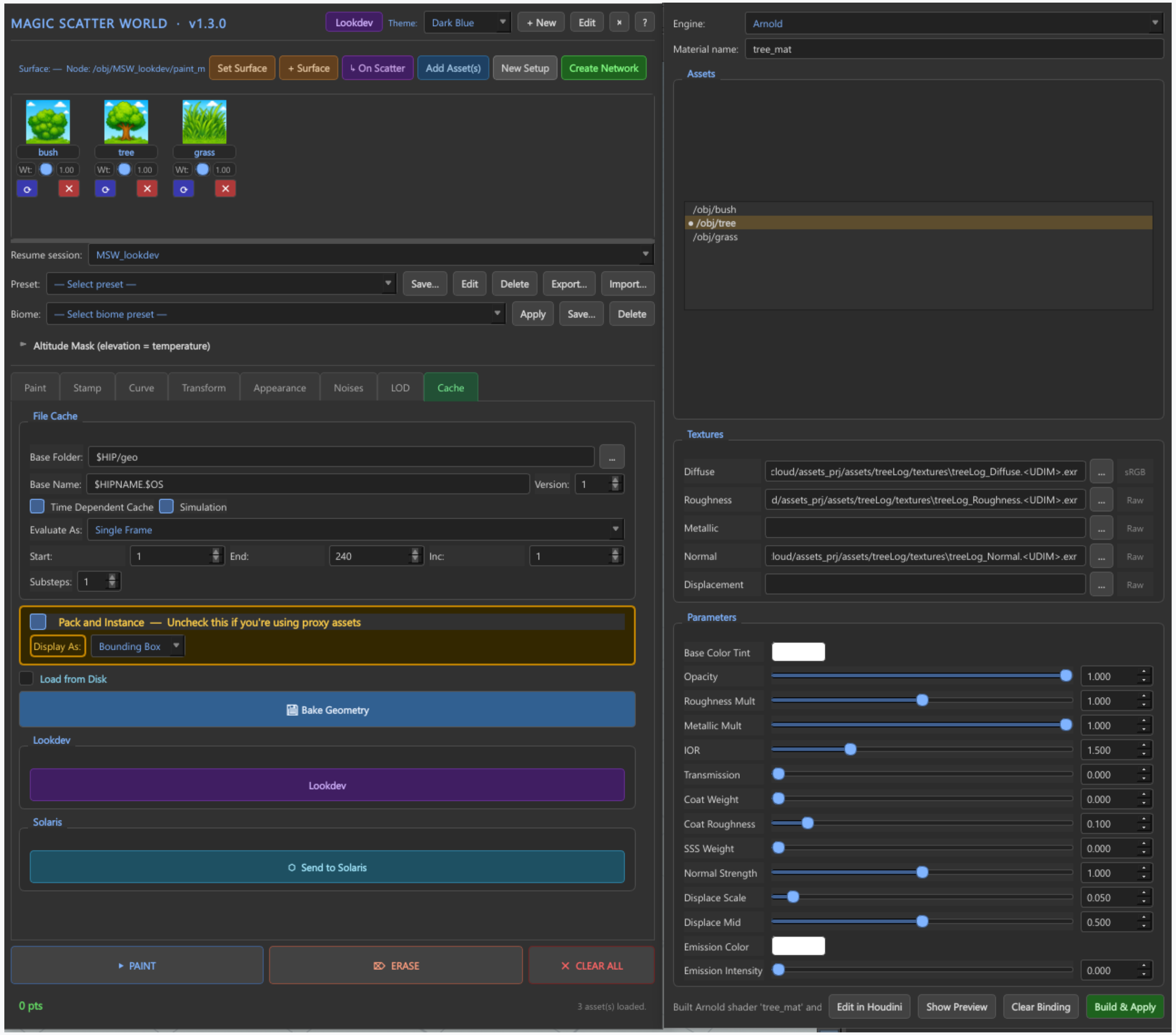
Lookdev

- Opens the non-modal Lookdev dialog for PBR material assignment
- Assign textures to Diffuse, Roughness, Metallic, Normal, Displacement slots
- Supports Arnold (HtoA) and Redshift native material builders
- Karma/USD via MaterialX with UDIM and explicit UV reader support

Solaris

- Include Wire Mesh — when enabled, wire geometry is included in the USD export
- Send to Solaris — exports ivy as USD PointInstancer into a Solaris LOP network
- Prototype USDs cached at $HIP/usd/prototypes/ with SHA1 invalidation

Lookdev · PBR material assignment · Arnold · Redshift · Karma / USD



Engine Selection

Arnold (HtoA) and Redshift native material network builders

Auto-detects the available renderer on tool launch

Texture Slots

5 slots: Diffuse · Roughness · Metallic · Normal · Displacement

Browse for any file path — supports UDIM glob patterns

Per-engine colorspace badges (sRGB / Raw) per slot

PBR Parameters

Roughness Mult — scale the roughness texture output

Metallic Mult — scale the metallic texture output

Normal Strength — intensity of the normal map

Coat Weight / Coat Roughness — clearcoat layer controls

SSS Weight — subsurface scattering amount

Emission Intensity / Emission Color — self-illumination

IOR — index of refraction for specular response

Transmission — glass and water transparency

Actions

Build & Bake — compile material network and assign to all assets

Show Painting — toggle asset visibility in viewport

Bake Geometry — freeze geometry with material assignments to disk

Send to Solaris — export with MaterialX shaders for Karma / USD

Quick Start Workflow · Grow your first ivy in minutes

1. Open from the Ivy Generation shelf button
2. Select your surface geometry → click Set Surface
3. Select ivy leaf/asset geometry → click Add Asset(s)
4. Click Create Network and name your setup
5. Paint Mask tab → paint where ivy should grow with PAINT
6. Ivy Generation tab → click Create Ivy Network
7. Adjust Strand Length, Step Size, Gravity for shape
8. Appearance tab — tweak Wire Radius and Noise settings
9. Transformation tab — add scale and rotation variation
10. Simulation tab → click Create Sim then Simulate for physics
11. Output/Bake tab → click Bake Geometry to freeze the result
12. Click Send to Solaris for USD / Karma rendering

All ivy data is saved automatically inside the .hip file — no sidecar files required.